

Predicting Pitch Sequences in Baseball Using Multinomial Logistic Regression

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Section 1

PART 1: NON-TECHNICAL

The Battle: Pitcher vs. Hitter

Every pitch is a decision:

- What type of pitch to throw?
- Where to throw it?
- What did I just throw? What's the count?

Why does it matter?

- A 95 mph fastball reaches home plate in ~**400 milliseconds**
- Knowing what's coming = precious extra reaction time
- The ability to anticipate a pitch can shift the competitive balance

The Core Question

"What pitch will the pitcher throw next?"

People have tried to answer this for a long time. From reading hand signals to high-tech scanning systems

Baseball Has a Memory

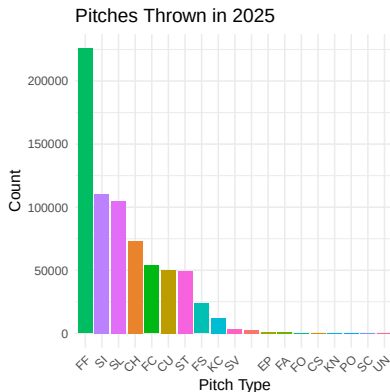
MLB's Statcast system tracks everything:

- Every pitch in every game, all season long
- **712,528 pitches** recorded in 2025
- Velocity, spin, location, outcome were all captured

Pitchers have habits:

- After a fastball, what comes next?
- Does the count change behavior?
- Does the batter's handedness matter?

Just like people have habits in daily life, pitchers develop tendencies over thousands of pitches.



The Approach: Building a Prediction Model

Inputs to the model:

- What were the last 1-2 pitches thrown?
- What is the current count (balls & strikes)?
- Does the batter bat left- or right-handed?

Output:

- A probability for each pitch type
- e.g., "40% *fastball*, 35% *slider*, 25% *changeup*"
- The model picks the **most likely** one as its prediction

Key design choice: I built a *separate* model for each of the 767 pitchers in 2025 - not one generic model

How to think about it

Situation

(count, last pitch, batter side)



Model learns pitcher's habits



Predicted next pitch probability

What Did I Find?

Prediction Accuracy:

- **62%** correct predictions vs a 32% random-guess baseline
- That's roughly **double** what you'd expect by chance

Key finding #1:

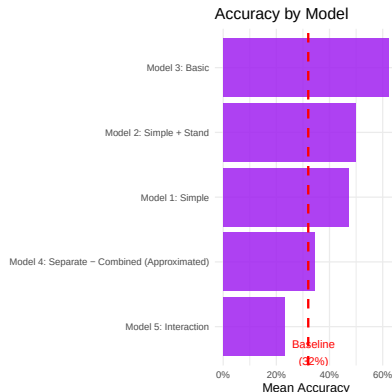
Relief pitchers (closers, setup men) are typically **more predictable** than starting pitchers

Key finding #2:

The **count** (balls & strikes) is one of the biggest drivers of what pitch comes next

Key finding #3:

Model complexity can backfire as more isn't always better



The App: Prediction in Real Time

The Shiny App puts the model to work:

- Select any 2025 MLB pitcher
- Enter the game situation (count, last pitch, batter side)
- Get probabilities for every pitch type

What the app shows you:

- Predicted pitch type and confidence level
- Full probability breakdown for all pitch types
- How accurate the model is for that specific pitcher

App Demo

Watch a **live video** of a pitcher and follow along pitch-by-pitch using the app.

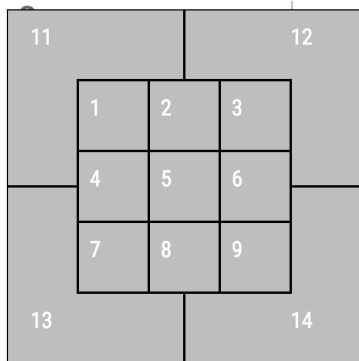
Practical Uses

- Pre-game scouting reports
- Hitter preparation
- Pitcher self-assessment

Access the app:

https://b-bruxvoort.shinyapps.io/pitch_type_prediction_app/

App Cheat Sheet: Zones & Pitch Types



Catcher/umpire's point of view

Pitch Type Abbreviations:

CH	Changeup
CU	Curveball
FC	Cutter
EP	Eephus
FF	Four-Seam Fastball
FO	Forkball
KN	Knuckleball
KC	Knuckle-curve
SC	Screwball
SI	Sinker
SL	Slider
SV	Slurve
FS	Splitter
ST	Sweeper

Questions?

Part 1 — General Overview

Thank you for your time and interest.

We will now transition into the technical presentation.

Section 2

PART 2: TECHNICAL DEEP DIVE

Research Objectives

Four core contributions:

- ➊ **Systematic model comparison** — Five multinomial logistic regression specifications of increasing complexity, evaluated against one another with formal statistical tests
- ➋ **Pitcher-specific models** — 767 individual models rather than a single league-wide model, capturing individual pitching tendencies
- ➌ **Practical variable restriction** — Only variables observable in real time during a live game; no spin axis, release height, or trajectory features
- ➍ **Interpretability emphasis** — Logistic regression over black-box alternatives, with coefficient interpretability as a design priority

Guiding question

Can classical statistical methods effectively model sequential pitch selection while remaining implementable in real-world game contexts?

Data: Statcast 2025

Source: MLB BaseballSavant

Scope: 2025 regular season, all 30 teams

Scale: 712,528 pitch observations, 118 raw variables

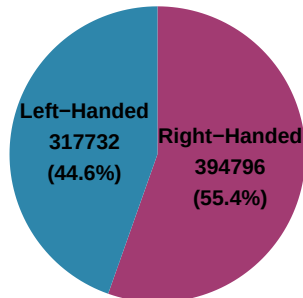
Variables retained:

- `pitch_type` — outcome variable (14 types)
- `zone` — pitch location (14 zones)
- `stand` — batter handedness (L/R)
- `balls`, `strikes` — count state
- `at_bat_number`, `pitch_number`, `player_name`

Why not spin rate, velocity, pfx?

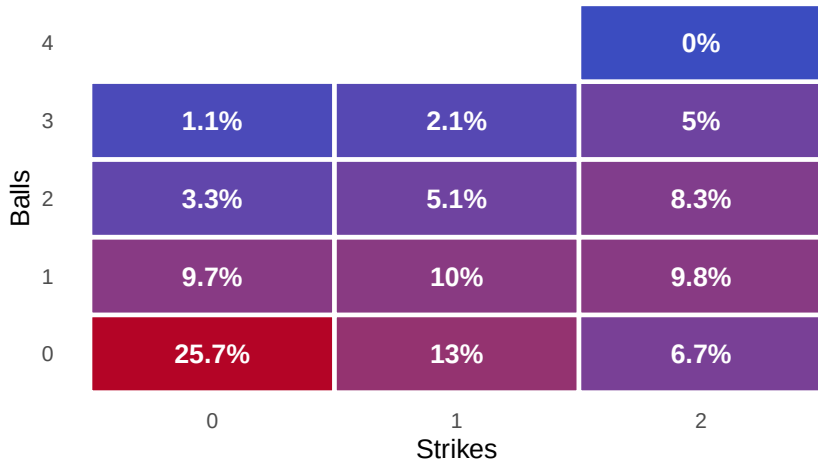
Multicollinearity, not real-time observable, overfitting risk

Batter Handedness Distribution



Exploratory Data Analysis: Count Distribution

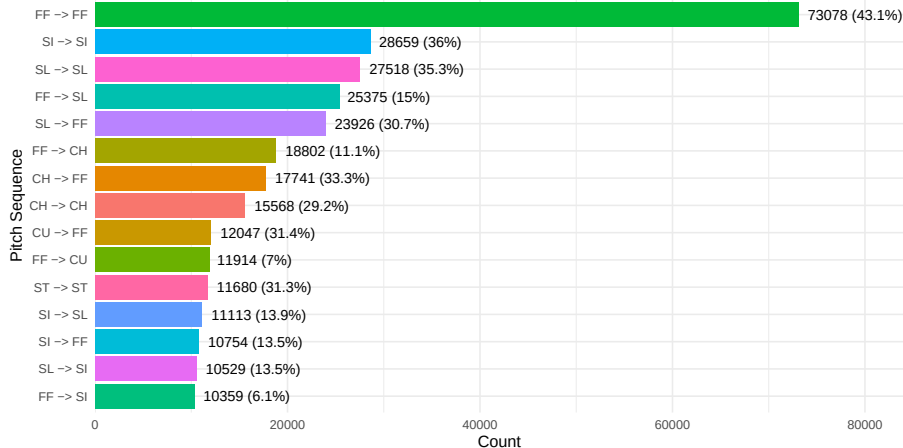
Count Distribution



Key takeaway: 0-0 count = 25.7% of all pitches. Every PA starts here, making it the single most common game state.

Exploratory Data Analysis: Pitch Sequences

Top 15 Most Common Pitch Sequences



Key takeaway: FF→FF occurs 43.1% of the time after a fastball as pitchers repeat their primary pitch most of the time.

Methodological Framework

Why Multinomial Logistic Regression?

For an outcome $Y_i \in \{1, \dots, J\}$ (pitch types), the softmax formulation:

$$P(Y_i = j \mid \mathbf{X}_i) = \frac{\exp(\mathbf{X}_i \beta_j)}{\sum_{k=1}^J \exp(\mathbf{X}_i \beta_k)}$$

Advantages:

- Natural fit for unordered categorical outcomes
- Coefficients interpretable as log-odds ratios
- Computationally efficient at scale
- Probabilities guaranteed to sum to 1

Why not tree-based / neural methods?

- Random forest and XGBoost tested but had no meaningful accuracy gain for computational power needed
- Pitch selection is approximately linear in observable game state
- Logistic regression is faster, interpretable, and deployable^{15 / 28}

The Key Addition: Lagged Variables

Pitch selection is inherently sequential. Pitchers “set up” batters across multiple pitches.

Lagged features created:

- `lag1_pitch_type` — previous pitch type
- `lag2_pitch_type` — two pitches ago
- `lag1_zone` — previous pitch location
- `lag2_zone` — location two pitches ago

At-bat boundary handling: Lag variables reset at the start of each at-bat. First pitch of a new PA receives "None" for all lags.

Model-building code (Model 3)

```
multinom(  
  pitch_type ~  
    prev_pitch_type_1 +  
    prev_zone_1 +  
    prev_pitch_type_2 +  
    prev_zone_2 +  
    balls + strikes + stand,  
  maxit = 500)
```

Fit per pitcher with ≥ 100 pitches. MLE via `nnet::multinom()`.

Five Model Specifications

Model	Predictors	Complexity
1: Simple	lag1/lag2 pitch type, pitch number	1
2: Simple + Stand	Model 1 + batter handedness	2
3: Basic (full)	Model 2 + balls, strikes, lag1/lag2 zone	4
4: Separate	Two models — type only & zone only	5
5: Interaction	Joint pitch type $\times$ zone outcome	8

Development order:

Model 3 built first (kitchen sink), then Models 4/5 tested alternative outcome structures, then Models 1/2 tested whether simpler approaches were competitive.

Model 5 rationale

Pitch type and location decisions are correlated, but can we predict them jointly as a single outcome (e.g., "slider in zone 14")?

Issue: this creates 100+ outcome categories and collapses under data sparsity.

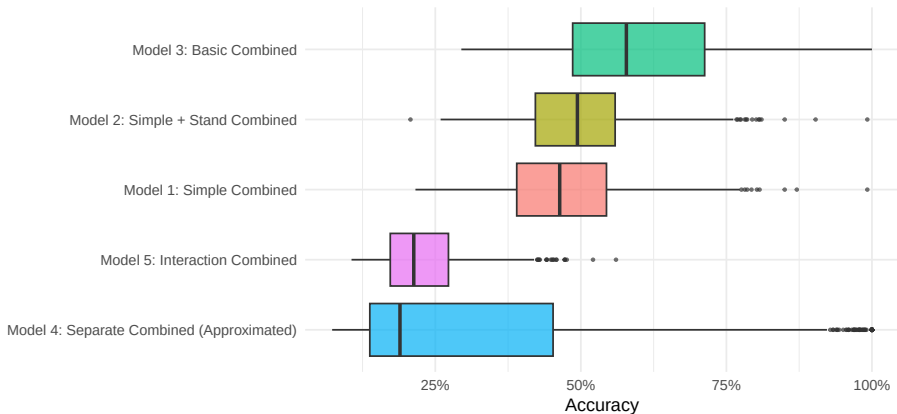
Overall Results: Model Performance (Summary)

Model	Mean	SD
Model 3: Basic	0.6236	0.1797
Model 2: Simple + Stand	0.4975	0.1070
Model 1: Simple	0.4720	0.1125
Model 4: Separate - Combined (Approximated)	0.3432	0.3044
Model 5: Interaction	0.2310	0.0811

Key: Model 3 = 62.36%; Model 1 = 47.20%; Model 5 = 23.10% (below baseline).

Overall Results: Model Performance (Distribution)

Accuracy Distribution by Model



Model 3's wide distribution signals strong pitcher-to-pitcher variation meaning it excels for predictable pitchers and struggles for unpredictable ones.

Statistical Significance Testing

ANOVA: $F(4, 3659) = 483.2$, $p < 2 \times 10^{-16}$ which shows overwhelming evidence of differences

Tukey HSD — Notable Pairwise Comparisons:

Comparison	Mean Diff	Cohen's d	Effect
M1 vs M2 (add handedness)	+2.54 pp	-0.232	Small
M1 vs M3 (add count+zone)	+15.16 pp	-1.011	Large
M2 vs M3 (isolate count+zone)	+12.62 pp	-0.853	Large
M3 vs M4 (joint prediction cost)	+28.04 pp	+1.122	Large
M3 vs M5 (interaction failure)	+39.26 pp	+3.413	Enormous
M4 vs M5 (separate vs joint)	+11.22 pp	+0.428	Medium

All 10 pairwise comparisons significant at $\alpha = 0.01$. Count and location contribute the largest marginal gains; Model 5 failure is structurally severe.

Why Model 5 Failed: Data Sparsity

The problem:

- 7–10 pitch types $\times$ 14 zones
= 98–140 outcome categories per pitcher
- A pitcher with 2,500 pitches may produce 80–100 unique type-zone combos
- Many combos appear only 5–10 times which is insufficient for stable MLE

Lesson: More information can hurt when complexity grows faster than data availability. The theoretical motivation was sound; the practical execution was not.

Why Model 4 fares better

Model 4 separates the two predictions. Each sub-model has a manageable number of outcome categories (14 pitch types *or* 14 zones). Combined accuracy is approximated as:

$$\hat{A}_{\text{combined}} = \hat{A}_{\text{type}} \times \hat{A}_{\text{zone}}$$

The multiplicative penalty still produces low combined accuracy (34.32%), but the individual models are statistically tractable.

Pitcher-Specific Analysis: The Relief vs. Starter Divide

Most predictable (min. 500 pitches):

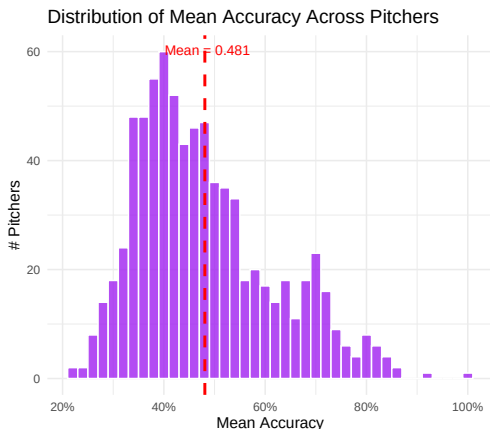
Pitcher	Acc	N
Kahnle, Tommy	0.703	1037
Hill, Tim	0.676	912
Jansen, Kenley	0.643	878
Mattson, Isaac	0.636	786
Pomeranz, Drew	0.630	805
Rogers, Tyler	0.616	978
Palencia, Daniel	0.592	845
Johnson, Pierce	0.591	979

Least predictable:

Pitcher	Acc	N
Lugo, Seth	0.226	2381
Darvish, Yu	0.229	1160
Lorenzen, Michael	0.235	2318
Buehler, Walker	0.243	2191
Martinez, Nick	0.254	2582
Alcantara, Sandy	0.255	2893
Quatrill, Cal	0.261	2006
Cabrera, Edward	0.261	2253

Pattern: Most predictable = relief pitchers (Kahnle, Jansen, Hill). Least predictable = experienced starters (Lugo, Darvish, Alcantara). **Elite starter unpredictability is a skill, not a modeling artifact.**

Pitcher Predictability Distribution



Key statistics:

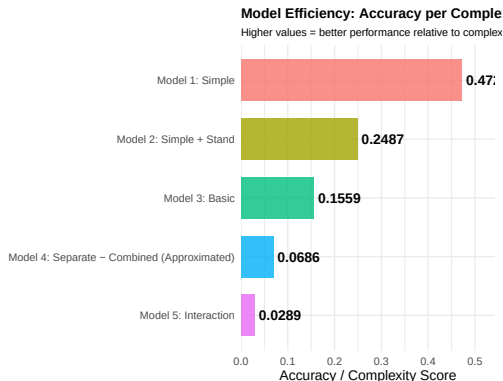
- Overall mean: **48.09%**
- Median: 45.60%
- SD: 13.53%
- Range: ~25% to 70%+

Accuracy–SD correlation:

$$r = 0.566 \ (p < 0.001)$$

More predictable pitchers show *greater model-to-model variance*. When real patterns exist, different models capture different aspects of those patterns. When no patterns exist, all models fail similarly.

Model Efficiency: Complexity vs. Accuracy



Recommendations by use case:

Max Accuracy

Model 3 — 62.36%, large effect sizes justify complexity

Best Efficiency

Model 1 — 47.20%, minimal computation, easy to deploy

Joint Type + Location

Model 4 (Separate) — only viable approach; 34.32%

Avoid

Model 5 — 23.10%, worst accuracy at highest complexity

Limitations

Data & temporal:

- Single season (2025) only meaning there was no temporal stability assessment
- Static models don't capture within-season adjustments (fatigue, injury, opponent adaptation)

Methodological:

- No formal cross-validation, but relative comparisons remain valid even if absolute accuracy is slightly optimistic

Contextual:

- Broad game context excluded: score differential, base-out state, inning, bullpen availability
- Catcher influence ignored. Catchers significantly shape pitch selection, especially for younger pitchers

Scope:

- No field experiment meaning we cannot establish that predictions improve offensive performance
- Results are correlational, not causal

Future Research Directions

Modeling improvements:

- **Multi-season analysis** — do pitcher patterns persist across years?
- **Hierarchical Bayesian models** — let pitchers "borrow strength" from similar pitchers
- **Recurrent neural networks** — longer sequence memory beyond a two-pitch lookback
- **Hidden Markov Models** — uncover latent pitcher "modes" (aggressive vs. conservative)

Substantive research questions:

- Does predictability correlate with ERA / WAR?
- Do patient hitters exploit predictable pitchers more than aggressive hitters?
- Can predictability changes serve as an early injury indicator?
- Can we add base-out state, score differential, inning, or more?

Possible Future Goal

Controlled field experiment where we deploy predictions to minor league hitters and measure offensive outcome changes

Conclusion

① **Game state drives selection.**

Count is the dominant predictor as adding it yielded a 12.62 pp gain ($d = -0.853$)

② **Role predicts predictability.**

Relief pitchers are more predictable; elite starter unpredictability is a cultivated skill

③ **Complexity must match data.**

Model 5's failure ($d = 3.413$) shows what happens when categories outgrow the data

④ **Classical methods are competitive.**

MLR at 62.36% matches neural network benchmarks while staying interpretable

Broader applicability

Categorical lagged variables in multinomial logistic regression can be applied beyond baseball where behavior unfolds in structured categorical sequences like healthcare, digital behavior, and other sequential decision systems.

Questions?

Part 2 — Technical Deep Dive

Thank you to Dr. Alberts and my committee
for their guidance throughout this research.

Full thesis, code, and app available upon request.

App: https://b-bruxvoort.shinyapps.io/pitch_type_prediction_app/